



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

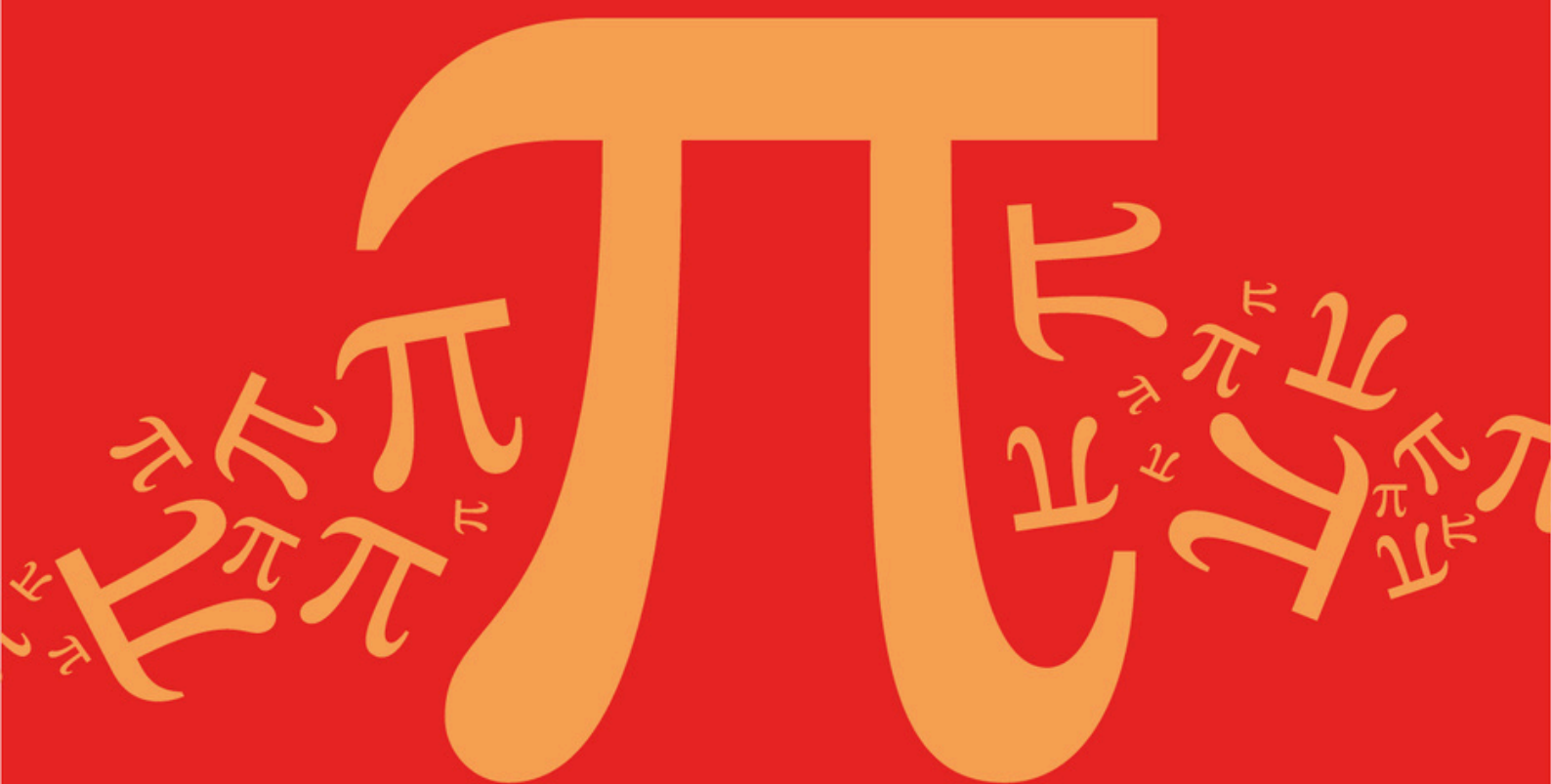
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB1: Terminology for Networks

Mark Scheme

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Mark Scheme

Topic

DB: Networks

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DB1: Terminology for Networks

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

- (a) (Odds B, D, F, H)

$$BD + FH = 37.2$$

$$BF + DH = 38.4$$

These 3 pairs of odds stated

M1

$$BH + DF = 40$$

3 correct totals, 2 correct totals

A2,1

$$\text{Length } 118 + 37.2$$

118 + their 'smallest' PI by their final answer

m1

$$= 155.2$$

CSO, including 3 correct totals.

A1

5

- (b) (i) E twice

B1

- (ii) I twice

B1

1

[7]

Q2.

- (a) 58

B1

1

- (b) $E A C D B E$

Or reverse

B1

1

- (c) $E A B D C E$

Tour

M1

(8 10 15 10 23)

Visit all vertices

m1

Correct order

A1

= 66

If M0 scored, then 66 scores SC2

A1

CSO

4

(d)

$\left. \begin{array}{l} AB \\ BD \\ DC \end{array} \right\} (35)$

A spanning tree with 3 edges connecting A, B, C and D and 2 edges from E

M1

Correct m s t

A1

$\left. \begin{array}{l} EA \\ EB \end{array} \right\} (17 = 52)$

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Correct edges from E

A1

52

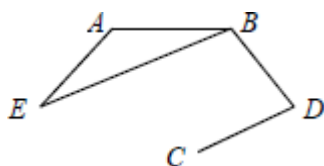
If M0 scored, then 52 scores SC2

A1

CSO

4

(e)



B1

Doesn't give a tour

Or other sensible conclusion

Eg: tour > 52 or 'doesn't give a solution'

E1
2
[12]

Q3.

(a) 103

B1
1

(b) Tour

May be improved

E1
E1
2

(c) A C D F B E A
(11 10 15 15 24 27)

Tour, from A, visiting at least 4 other vertices, once only

M1

Visits all vertices

m1

Correct order

= 102

If M0 scored then 102 scores SC2

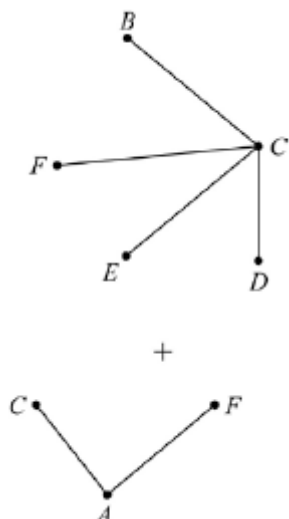
A1

A1

CSO

4

(d)



Spanning tree connecting B, C, D, E, F
 AND
 2 labelled edges from A
 (for both, edges, not lengths, can be either
 listed or shown in diagram)

Correct ST

Correct edges from A

M1

A1

A1

= 77

If M0 scored then 77 scores SC2

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A1

CSO

4

(e) Min tour ≥ 77

Allow their '77', provided '77' > 75

E1

1

[12]

Q4.

(a) (Odds A, C, L, G)

$$AC + LG = 27$$

$$AL + CG = 26$$

$$AG + CL = 30$$

These 3 sets of pairs stated

M1

One mark for each correct total

A1 x 3

Min 134 + 26

134 + their min of 3 totals.

m1

= 160

Must have scored first 5 marks.

A1

If M0 scored, then 160 scores SC2

CSO

6

(b) 4

B1

1

[7]

Q5.

(a)

$$\begin{array}{lcl} BD+FH & = & \left[\begin{array}{c} 210+210 \\ 200+180 \\ 260+340 \end{array} \right] \begin{array}{l} = 420 \\ = 380 \\ = 600 \end{array} \\ BF+DH & = & \\ BH+DF & = & \end{array}$$

These 3 sets of pairs

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M1

3 correct totals, 2 correct totals

A2,1

(MIN) = 2430 + 380

2430 + their smallest of three pair totals

m1

= 2810

CSO

A1

5

(b) 2430 + 340 (DF) = 2770

2430 + their DF

B1F

1

- (c) (i) $2430 + 180 (DH) = 2610$
 $2430 + \text{their min (must have scored M1)}$

B1F

1

- (ii) B, F only

B1

1

[8]

Q6.

- (a) Min MST

4 edges

M1

$$= 8 + 10 + 10 + 11 = 39$$

A1

2

- (b) Max MST = $8 + 17 + 17 + 18$
 $8 + 18 + 2 \text{ others}$

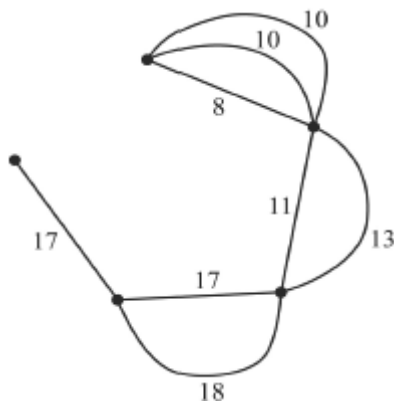
M1

$$= 60$$

A1

2

- (c)



Connected graph with 5 vertices (all edges numbered, from G)

M1

$$\text{MST} = 53$$

$$8, 11, 17, 17 \text{ or } 8, 10, 17, 18$$

A1

other edges OE

A1

(other possibilities not shown)
(all edges numbered, from G)

3

[7]

Q7.

(a) (i) 9

B1

1

(ii) $n - 1$

B1

1

(b) (i)

EF	8
BC	8.5
CG	10
JI	11.5
BI	12
AB	14
GE	16
CH	16.5
DE	21

SCA minimum spanning tree, 7+ **edges**
(not cycles), must be in ascending order
and edges required (not lengths alone)

M1

BC 2nd

A1

JI 4th

A1

9 edges (not lengths alone) – may be
earned in (b)(iii)

B1

All correct

A1

5

(ii) 117.5

B1

1

(iii)



7+ edges, minimum spanning tree

M1

Correct, including labelling

A1

2

[10]

Q8.

(a) (i) $(ABDK) = 3$

B1

1

(ii) $MST = (n - 1)$

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M1

= 15

A1

2

(b) (i) Odd vertices A and P

Route length = all edges + repeated edges (answer > 30)

M1 M1

Route length = all + (3)

A1

$(33 + 3) = 36$

B1

4

(ii) Repeated: AE, EL, LP

B1

1

(iii) E and L now odd

M1

$$\text{Their}(33) - 2 + 1 = 32$$

A1F

2

[10]

Q9.

(a) $3x + 6 < x + 24$

B1

$$3x + 6 < 2x + 17$$

(-1 for =)

B1

$$3x + 6 < 5x$$

SC J → M can score B2 (ie - 1 MR)

B1

3

(b) $x = 4 < (2x - 3)$

M1 A1

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$$x + 4 < (3x - 2)$$

OE

A1

3

(c) From (a)

$$x > 3$$

B1

$$x < 9 \text{ B1}$$

From (b)

$$x > 7$$

(2x > 6 ignore)

B1

$$7 < x < 9$$

$$x = 8$$

Tour 104

B1

B1

5

[11]

Q10.

(a) 3

B1

1

(b) (i) $n - 1$

B1

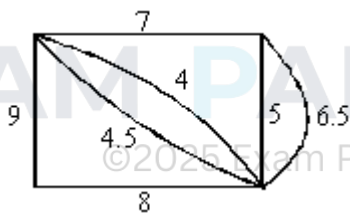
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(ii) $(n - 1)!$
M1 for factorial

M1A1

2

(c)



connected graph

M1

$m.s.t. = 17$

M1

all other arcs used and correct

A1

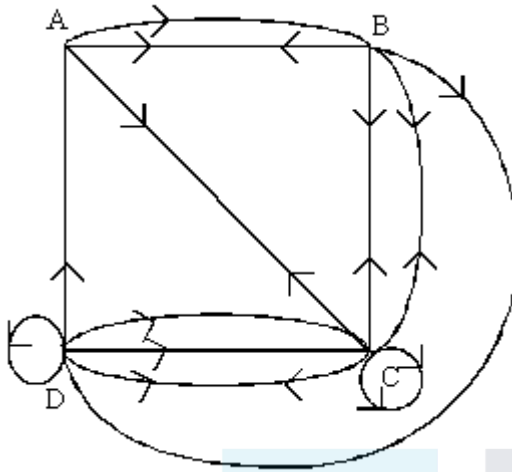
A1

3

[7]

Q11.

(a)



Directed graph with one correct

$C \rightarrow C, D \rightarrow$ correct

Fully correct

B1

B1

B1

3

(b) Not symmetrical

OE

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E1

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[4]

Examiner reports

Q1.

Generally the answers to part (a) were a very high scoring. Working was shown clearly and calculated accurately although there was, as in previous papers, a surprising number of students who didn't believe that the total length of the network was 118! The second part caused more difficulties. Many students did not realise that 'the number of visits' = 'degree of vertex' / 2, and resorted to finding a detailed route, with varying degrees of success, and then counting each occurrence of a vertex.

Q2.

Several students ignored the paper rubric and presented their working on the diagram in the question. The first three parts of the question were usually answered very well. Most knew the required algorithm for part (d), but while presentation is now significantly improved from that of past years, a significant minority still lost marks by not clearly detailing which edges were involved. Lettered labelling is required, not just numbers. For the last part, those who had the required two parts for the diagram remembered to join them. Surprisingly few students realised that the fact that this diagram was not that of a tour was significant, with many students stating that the route was the optimal solution (presumably because that is what the comment often involves)!

Q3.

A significant number of students failed to read the instructions on the title page and within every designated answer space, and used the diagram given in the text of the question.

- (a) This part was almost invariably answered correctly.
- (b) This was the first question to cause significant difficulty. The most common misconception was to believe that the answer in the question could certainly be improved. Many students failed to state the obvious but necessary fact that the answer was in fact the time for a tour.
- (c) This part was usually answered correctly. However, a common error was to fail to complete the tour by returning to *A*. Those who used a redrawn table often failed to make clear the order in which edges were to be chosen.
- (d) The method is known and understood by most students. However, there is a widespread tendency to deal merely in numbers and to not clearly identify the edges being used. This results in the loss of a significant number of marks. Many students thought they needed a tour of *BCDEF*.
- (e) The reason was apparently widely known but the ability to express it clearly in comprehensible words was often lacking.

Q4.

The method required for (a) was widely known, clearly explained and accurately calculated. The most common errors were to identify three or five odd order nodes. The incorrect answer of 3 was a frequently seen answer to part (b), obtained by dividing 6 by 2 and ignoring the correct method applied in the previous part.

Q5.

Part (a) was well-answered – students had been well-drilled in the rote use of an elementary algorithm, although arithmetical accuracy was sometimes weak. However, in the remainder of the question, students were required to do a little elementary analysis and thinking and many failed completely. It appears that these modified situations are not handled well. Students need to spend time studying semi-Eulerian situations.

Q6.

The usual mark scored was 4 out of 7, with very few candidates getting part (b) or the final mark in part (c) due to adding in extra edges and changing the MST.

Part (a) was usually correct.

Part (b) was usually incorrect, with $13 + 17 + 17 + 18 = 65$ being the almost universal answer. The very few who answered correctly were by no means always drawn from those scoring highest on the rest of the paper.

In part (c), most of the candidature spotted one of the two correct MSTs. A number of candidates elected to draw only a correct MST. Those candidates who attempted to complete the diagram frequently forfeited marks as their extra edges impacted on their MST. A few candidates drew separate diagrams.

Q7.

Part (a)(i) was almost always correct; likewise for part (a)(ii), although a number of candidates gave their answers as 11 and $n + 1$. Part (b) was usually correct, apart from the occasional slip with the 8th edge. Very few candidates failed to make clear their method at all. They were probably equal in number to those clearly using another method.

Q8.

Part (a) was well answered.

Part (b) proved to be more challenging. Candidates were able to make a good attempt at part (i), although a significant number used $16 + 2$ or 3. In part (ii), a common mistake was to omit the edge EL . Part (iii) proved to be a good discriminator. Candidates failed to realise that by removing AE and LP , then E and L would become the odd vertices.

Q9.

Although this question was an algebraic application of a travelling salesman problem, candidates were led through the question in stages. Nearly all candidates scored some marks on this question and the candidates who followed the instructions carefully scored full marks.

Q10.

Most candidates made a reasonable attempt at this question but very few scored full marks. Many did not see the connection between parts (a) and (b)(i). In part (c) many candidates were able to draw a correct minimum spanning tree but very few were able to add on the remaining arcs successfully.

Q11.

This question was a straightforward graph theory question. Unfortunately, a number of candidates thought that the numbers represented lengths and they then stated that the graph must be directed if the distance from A to B was different to the distance from B to A .



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